

Supply Transport Robot for Planetary Missions (STR-PM): Design and Implementation of an Introductory Autonomous Logistics System

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Abstract— This paper presents the design, development, and evaluation of a Supply Transport Robot for Planetary Missions (STR-PM), an introductory autonomous robotic system intended to simulate logistics operations in extraterrestrial environments. The system is designed to address fundamental challenges associated with space-based supply transport, including limited energy resources, environmental uncertainty, and the need for autonomous decision-making. The robot integrates an Arduino-based control architecture with ultrasonic and infrared sensing modalities to enable obstacle avoidance and line-following behaviors. A lightweight, energy-efficient mechanical structure is employed to reflect constraints typical of planetary missions. The project follows a structured engineering workflow consisting of system design, hardware integration, embedded programming, and experimental validation. While simplified for educational purposes, the STR-PM demonstrates foundational principles of autonomous robotics, including sensor-driven control, system reliability, and operational efficiency, and serves as a conceptual steppingstone toward more advanced robotic logistics systems for space exploration.

I. INTRODUCTION

Space exploration missions require reliable and efficient logistics systems to support astronaut operations and sustain mission continuity. Transporting supplies across planetary surfaces or orbital platforms presents significant

challenges due to environmental hazards, communication delays, and energy limitations.

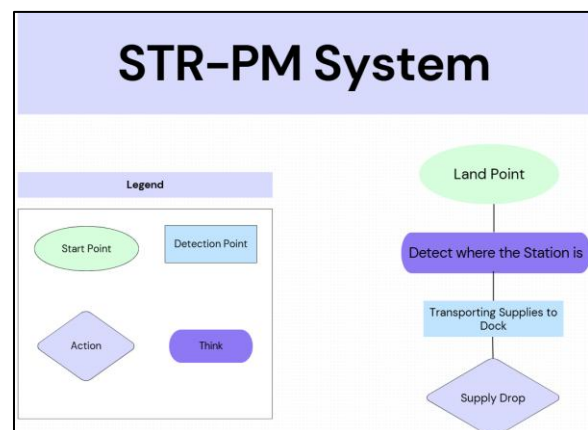
Robotic systems offer a viable solution by enabling autonomous or semi-autonomous transport of materials in environments that are hazardous or inaccessible to humans. This paper introduces the Supply Transport Robot for Planetary Missions (STR-PM), a ground-based prototype designed to simulate robotic logistics operations in extraterrestrial settings such as the Moon or Mars. The objective of this project is not to replicate full-scale space robotics systems, but rather to model key operational principles—namely autonomy, efficiency, and reliability—within a simplified, educational framework.

II. SYSTEM OVERVIEW

The STR-PM is a mobile robotic platform designed to perform autonomous navigation and simulated supply delivery tasks. The system follows a sequential operational flow:

1. Initialization and deployment
2. Environment sensing
3. Navigation and obstacle avoidance
4. Supply transport and delivery
5. Post-mission behavior (return, standby, or reassignment)

As illustrated conceptually in the flowchart, the robot detects a target location, navigates toward it, and performs a supply drop operation before transitioning to a post-mission state.



(Figure 1 – System Behavior)

III. DESIGN METHODOLOGY

The development of the STR-PM follows a structured engineering workflow:

A. Application Definition

The system is designed to simulate supply transport in space missions, emphasizing logistics support and autonomous operation.

B. Mechanical Design

A lightweight chassis is used to emulate mass constraints typical of aerospace systems. The robot employs a four-wheel differential drive configuration for mobility and stability.

C. Hardware Selection

Component selection prioritizes simplicity, reliability, and energy efficiency to reflect real-world engineering trade-offs.

D. Embedded Programming

The robot is programmed in C/C++ using an Arduino microcontroller, enabling low-level hardware control and efficient resource management.

E. Testing and Validation

Simulation and physical testing are conducted to evaluate system performance under controlled conditions, ensuring functional reliability.

IV. SYSTEM ARCHITECTURE

A. Hardware Components

The system consists of the following subsystems:

- **Mobility System:**
 - DC gear motors
 - Wheels and chassis
- **Control System:**
 - Arduino UNO microcontroller
 - Motor driver (L298N)
- **Sensing System:**
 - Ultrasonic sensor (distance measurement)
 - Infrared sensors (line tracking)
- **Actuation System:**

- Servo motor for payload interaction

- **Power System:**

- Lithium battery pack

B. Software Architecture

The control logic is based on sensor-driven decision-making:

- **Obstacle Avoidance:**
Ultrasonic sensor detects objects and adjusts motion accordingly.
- **Line Following:**
IR sensors enable path tracking for guided navigation.
- **Task Execution:**
The robot performs predefined behaviors such as movement, stopping, and simulated payload deployment.

This modular approach reflects fundamental principles used in autonomous robotic systems.

V. ENGINEERING CONSTRAINTS AND LIMITATIONS

A. Environmental Constraints

Space environments introduce extreme temperature variations, vacuum conditions, and radiation exposure. While not physically replicated, these constraints inform design decisions emphasizing durability and simplicity.

B. Energy Constraints

Limited power availability necessitates efficient system design. The use of low-power components reflects this requirement.

C. Communication Constraints

Due to signal delays in space, robots must operate autonomously. The STR-PM simulates this through onboard decision-making capabilities.

D. Payload Limitations

Physical constraints of the chassis limit payload capacity, highlighting trade-offs between functionality and system size.

VI. TESTING AND RESULTS

Testing focused on validating core functionalities:

- Obstacle detection and avoidance
- Line-following navigation
- Basic autonomous movement

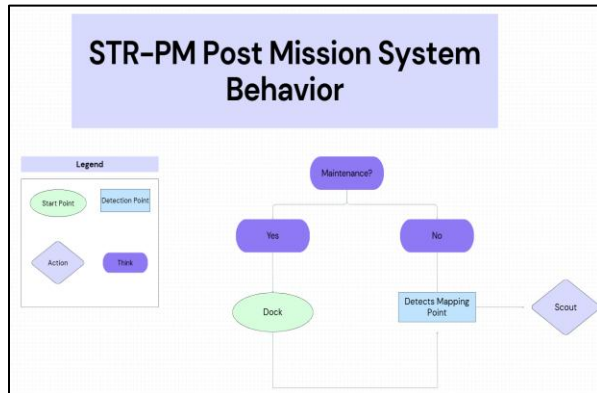
Results indicate that the system successfully performs these tasks under controlled conditions. While performance is limited by sensor precision and processing capability, the robot demonstrates reliable operation within its intended scope.

VII. POST-MISSION BEHAVIOR

After completing its primary task, the STR-PM can transition into different operational states:

- Return-to-base for recharging
- Standby or monitoring mode
- Reassignment to secondary tasks

As shown in the system flowchart, this introduces the concept of autonomous mission lifecycle management.



(Figure 2 – Post Mission System Behavior)

VIII. DISCUSSION

The STR-PM highlights several key principles relevant to space robotics:

- Autonomy: Sensor-based decision-making enables independent operation
- Efficiency: Lightweight and low-power design aligns with mission constraints

- Scalability: The system can be extended with more advanced sensors and control algorithms

Although simplified, the project provides a foundational understanding of how robotic systems can support logistics in space exploration.

IX. CONCLUSION

This paper presented the design and implementation of the STR-PM, an introductory robotic system for simulating supply transport in planetary missions. The project demonstrates core concepts in autonomous robotics, including navigation, sensing, and system integration, within a constrained and educational framework.

Future work may include:

- Integration of advanced sensing (e.g., vision systems or LiDAR)
- Implementation of adaptive path planning algorithms and exploration of multi-robot coordination.

The STR-PM serves as a steppingstone toward more complex and mission-relevant robotic systems.